Incident Report: Malware Execution and Command-and-Control Communication

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Alert Title

Suspicious PowerShell Execution with External Connection

Summary

A suspicious process execution was observed on the host `DESKTOP-92A1` involving `powershell.exe` launched via an unusual parent process (`winword.exe`). The PowerShell command was encoded, indicating obfuscation. This activity was followed by an outbound network connection to an external IP address over HTTPS, along with high-volume data transfer. Additionally, a suspicious executable file was dropped in the user's downloads directory, and an antivirus alert was triggered. The behavior strongly indicates malware execution with potential command-and-control (C2) communication.

Initial Findings

- User: `faisal.khan`
- Host: `DESKTOP-92A1`
- Time: 11:45 PM
- Process: `powershell.exe`
- Command: Encoded / Obfuscated
- Parent Process: `winword.exe`
- Outbound IP: `185.225.69.47`
- Port: 443
- Additional: Suspicious executable dropped in Downloads directory

Key Observations

- Execution during non-working hours (11:45 PM)
- PowerShell launched with encoded command (obfuscation technique)
- Unusual parent-child relationship (Word spawning PowerShell)
- Outbound connection to external infrastructure (likely VPS)
- High-volume data transfer indicating potential data exfiltration

- Suspicious file dropped and flagged by antivirus

Investigation

Suspicious Process Execution

```
DeviceProcessEvents
| where FileName == "powershell.exe"
| where ProcessCommandLine has_any ("-enc", "EncodedCommand")
| summarize ExecutionCount = count()
    by DeviceName, AccountName, ProcessCommandLine, bin(TimeGenerated, 10m)
| sort by ExecutionCount desc
```

Result: Confirmed suspicious PowerShell execution with encoded commands.

Network Activity (Conceptual)

- Identified outbound connection to external IP (`185.225.69.47`)
- Observed high data transfer volume over HTTPS
- Behavior consistent with command-and-control communication

Evidence

- PowerShell execution with encoded command
- Parent process: `winword.exe`
- Outbound connection to external IP
- High-volume data transfer
- Suspicious executable file detected
- Antivirus alert triggered

Conclusion

This incident is classified as a **Confirmed True Positive (Malware Execution with Possible C2 Communication)**.
The attack likely originated from a malicious document leading to PowerShell-based execution and external communication.

Severity

Critical

Response Actions

- Isolated the affected host from the network
- Terminated suspicious PowerShell process
- Removed malicious executable file
- Blocked outbound connection to malicious IP
- User account temporarily restricted
- Incident escalated to Tier 2 for further investigation

Analyst Note

This incident highlights the risks associated with malicious document-based attacks and the importance of monitoring parent-child process relationships. Detection rules should be enhanced to flag suspicious PowerShell executions, especially those spawned by Office applications, along with outbound connections to unknown external infrastructure.